



CAS

IEEE CIRCUITS AND SYSTEMS SOCIETY

UNIC-CASS

Mock Tapeout 2

“250 MHz *Quadrature Delay Locked Loop*”

27.01.2026



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3. **Schematics and Layout:** TOP and Sub-blocks: Symbols & Schematics & Layouts
4. **QDLL TOP Layout Verification:** Top Layout and layout within the pad ring.
5. **QDLL:** Pins and IO Cells.
6. **Project Status**

2nd UNIC-CASS Progress Questionnaire

Progress Questionnaire

Design Team Identifier (Use an ACRONYM as a short name for the IP block top level) *

QDLL

Primary contact email *

vazquezleonardodavid@outlook.com

Secondary contact email *

leonardo.vazquez@ieee.org

Active Design Team members (First Name & Last Name of all participants, except mentor) *

Leonardo David Vazquez
Santiago Basignana
Mateo Buteler
Clemente Molinari

Mentor's Name(s) *

Agustín Galetto

Repository link (e.g. <https://github.com/username/repository-name>) *

https://github.com/leonardovazquez/unic-cass-wrapper_QDLL
[leonardovazquez/unic-cass-wrapper_QDLL: Unic-Cass Repo for the analog project "QDLL" in 130nm sg13g2 tech](#)

Link to tapeout report (provide link to e.g. Google docs file with your steps and outputs) *

<https://docs.google.com/presentation/d/1SaUPbj3oBROP9z1-2TtK8tWlXL02-J9qEyfox9xo-il/edit?usp=sharing>

<https://docs.google.com/presentation/d/1SaUPbj3oBROP9z1-2TtK8tWlXL02-J9qEyfox9xo-il/edit?usp=sharing>

Design Team Country *

ARGENTINA ▼

Progress Questionnaire

Is your design? *

- ☐ Digital Only
- ☒ Analog / Mixed Signal

Do you have special I/O requirements (RF Signal, High Voltage, High Current, low R/C Parasitics, low leakage)?

If so, please explain with as much details as possible

The QDLL project basically consist on a single block with 1 differential input signal (2 inputs) and 1 differential output signal (2 outputs). For a CMOS (0-1.2 V) 250•MHz input signal, standard IO cells have been tested and found to perform well. However, if available, we prefer to interface with RF 50• Ω systems using IO cells with matched impedance. We would need the new RF iocells models. If this is not possible, as we have demonstrated, we will use the IO Cells sg13g2_IOPADAnalog that we mentioned in our document. Also, we can migrate our project to a lower frequency, there is no problem. Thanks!

For Analog / Mixed-Signal Design

Is the circuit schematic for your Design already completed in Xschem? *

- ☒ Yes
- ☐ No - Explain below

If the schematic is not completed, tell us where you are at

Tu respuesta

What is your Core Area in mm² (without the pads!) *

- ☒ < 0.1 mm²
- ☐ 0.1 mm² to 0.2 mm²
- ☐ 0.2 mm² to 0.3 mm²
- ☐ 0.3 mm² to 0.4 mm²
- ☐ 0.4 mm² to 0.5 mm²
- ☐ 0.6 mm² to 0.7 mm²
- ☐ 0.7 mm² to 0.8 mm²
- ☐ 0.8 mm² to 0.9 mm²
- ☐ 0.9 mm² to 1 mm²
- ☐ > 1 mm²

Progress Questionnaire

What is your Core Area in mm² (without the pads!) *

- ☒ < 0.1 mm²
- ☐ 0.1 mm² to 0.2 mm²
- ☐ 0.2 mm² to 0.3 mm²
- ☐ 0.3 mm² to 0.4 mm²
- ☐ 0.4 mm² to 0.5 mm²
- ☐ 0.6 mm² to 0.7 mm²
- ☐ 0.7 mm² to 0.8 mm²
- ☐ 0.8 mm² to 0.9 mm²
- ☐ 0.9 mm² to 1 mm²
- ☐ > 1 mm²

How many VDD Pads (0-2) *

1

Which voltage option are you using as your supply voltage (VDD): *

- ☒ 1.2V
- ☐ 3.3V

What is the expected current requirement for you supply pads (e.g., 0.1mA)

The average current on typical operation is 0.45 mA.
We preferred to use the 4mA supply pads

How many VSS Pads (0-2) *

1

How many IOVDD Pads (0-2) *

1

How many IOVSS Pads (0-2) *

1

How many Analog Pads (0-18) *

4

Have you completed the layout of what percentage of your full custom Design *
(choose from 0 = 0% to 10 = 100%)

Nothing ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☒ 10 Complete Layout

Progress Questionnaire

Have you completed the layout of what percentage of your full custom Design *
(choose from 0 = 0% to 10 = 100%)

Nothing ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☒ 10 Complete Layout

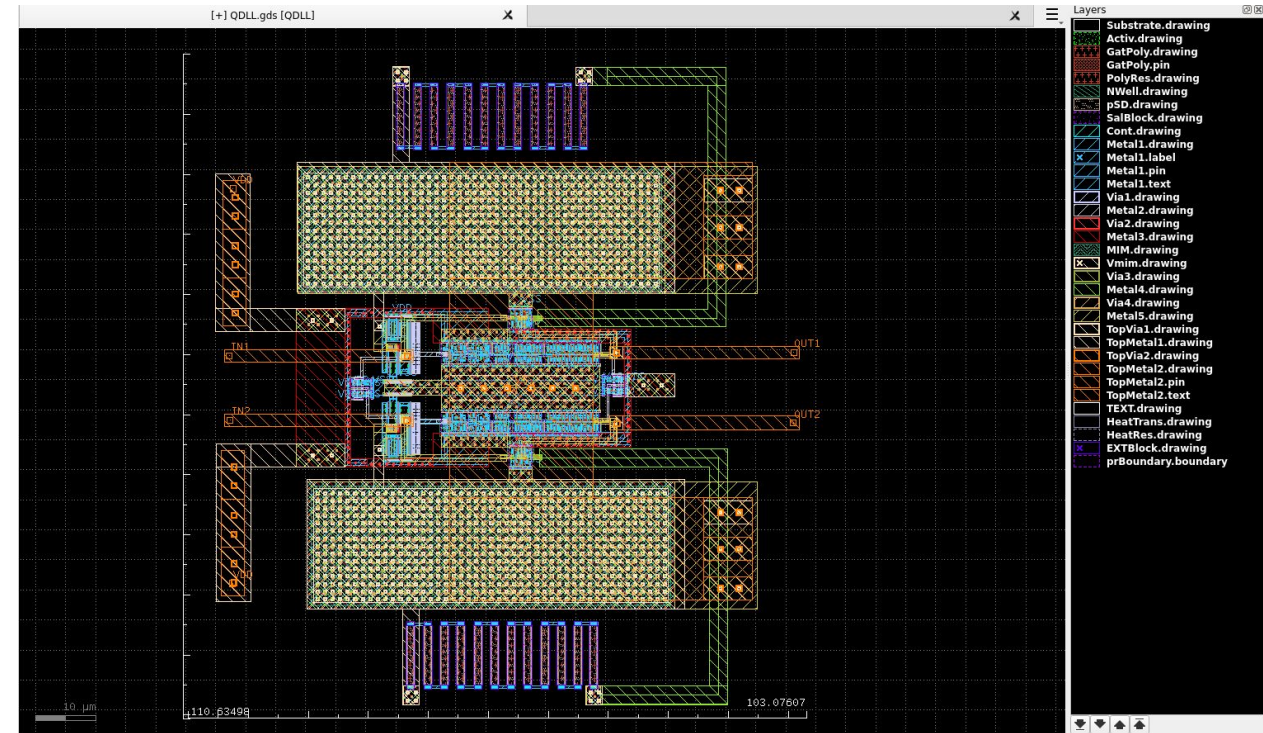
The digital blocks in your design, if needed, are they completed in RTL?

- ☐ Yes
- ☐ No
- ☒ Don't apply (fully analog)

Borrar la selección

What parts or blocks of your full custom Design are already laid-out and DRC & LVS verified with KLayout?

All blocks and the TOP have passed DRC and LVS verification with KLayout



Progress Questionnaire

Write two sentences about the status of your Design and the design team familiarity with the Analog Open-source EDA flow.

Most members of the design team are graduate and undergraduate students with less than one year of experience using analog open-source EDA tools. Nevertheless, we have become well familiarized with the design flow, supported by the fact that two team members completed an open-source analog design course taught by Prof. Jorge Marín within the UNICCAS program, which has been very beneficial for transferring knowledge to the rest of the team.

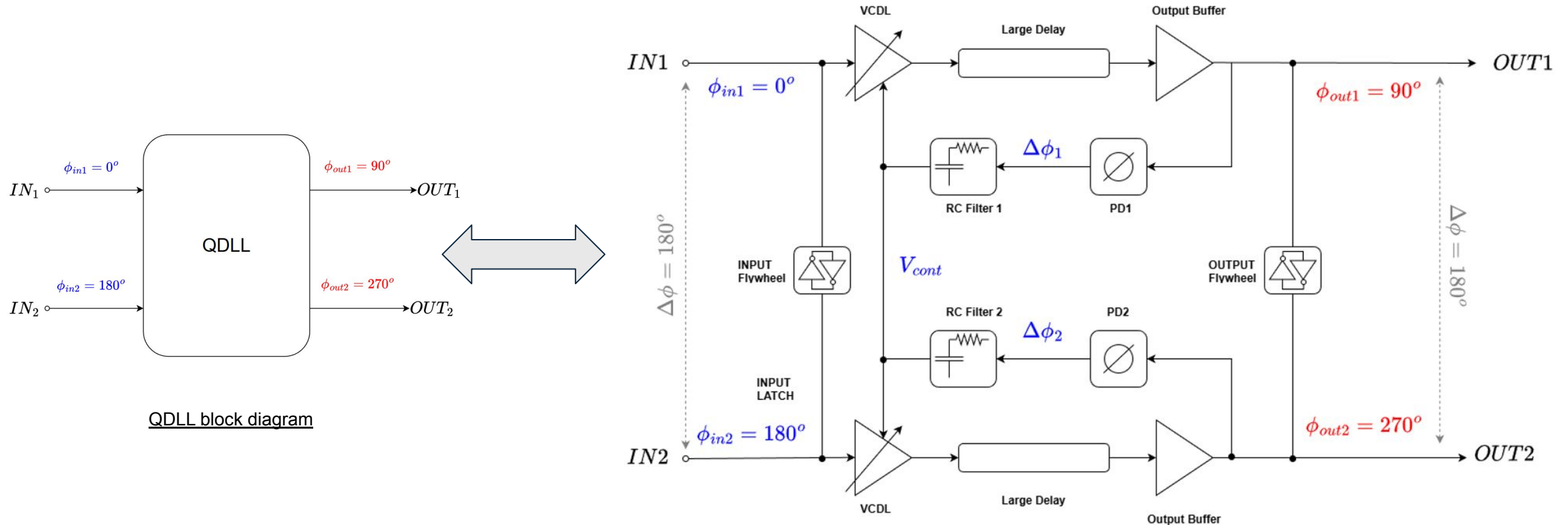
Write two sentences about difficulties, if any, the design team encountered with the Open-source tools and difficulties, PDK usage.

We encountered issues related to tool versions, as the UNICCAS repository does not provide the latest releases of Magic and KLayout. However, we have been able to work around these limitations and continue making progress in our project.

2- QDLL Project Diagram & Specifications

2 - QDLL Project Diagram

For this project, two single-ended DLLs with differential outputs are used to generate two differential signals with 90° and 270° phase shifts. The RC filters and the large VCDLs are designed to stabilize each branch at a $90^\circ \pm 5^\circ$ phase difference at **250 MHz**. To ensure synchronization between the signal pairs, flywheel cells are added.



2 - QDLL Project Specifications

The design specifications were modified due to the high-frequency behavior of the I/O cells. As a result, the performance requirements were relaxed, along with adjustments to the project focus and the number of input/output pins.

Pins:

- 2 Bi-directional: VDD, VSS
- 2 inputs: inpad1, inpad2
- 2 outputs: outpad1, outpad2

Estimated Number of Pins:*

Input: *

2

Output: *

2

Bidirectional: *

2

Design Type:**

Analog



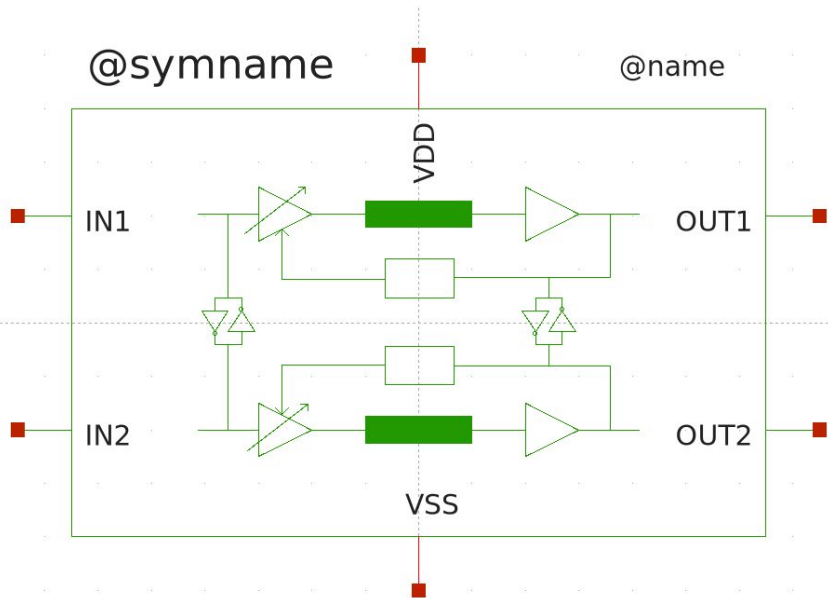
ihp 130 nm

QDLL Specifications	
Boundary Conditions	Requirement
Techn.	CMOS
Node	130 nm
Supply Voltage	1.2 ± 5% V
Temperature	[0, 65, 125] °C
Input Amplitude	[0 , 1.2] V (CMOS)
Input Frequency	250±25 MHz
Input/Output Type	Fully Differential
Specifications	Value
Phase Difference	90 ± 5 degrees
Output Swing	[0 , 1.2] V (CMOS)
t _{rise} /t _{fall} output	<
V _{cont_ripple}	<
Output Jitter	-

3- QDLL Project Schematics & Layouts

3 - QDLL Schematics (TOP)

- Differential QDLL TOP:



QDLL.sym

Estimated Number of Pins:*

Input: *

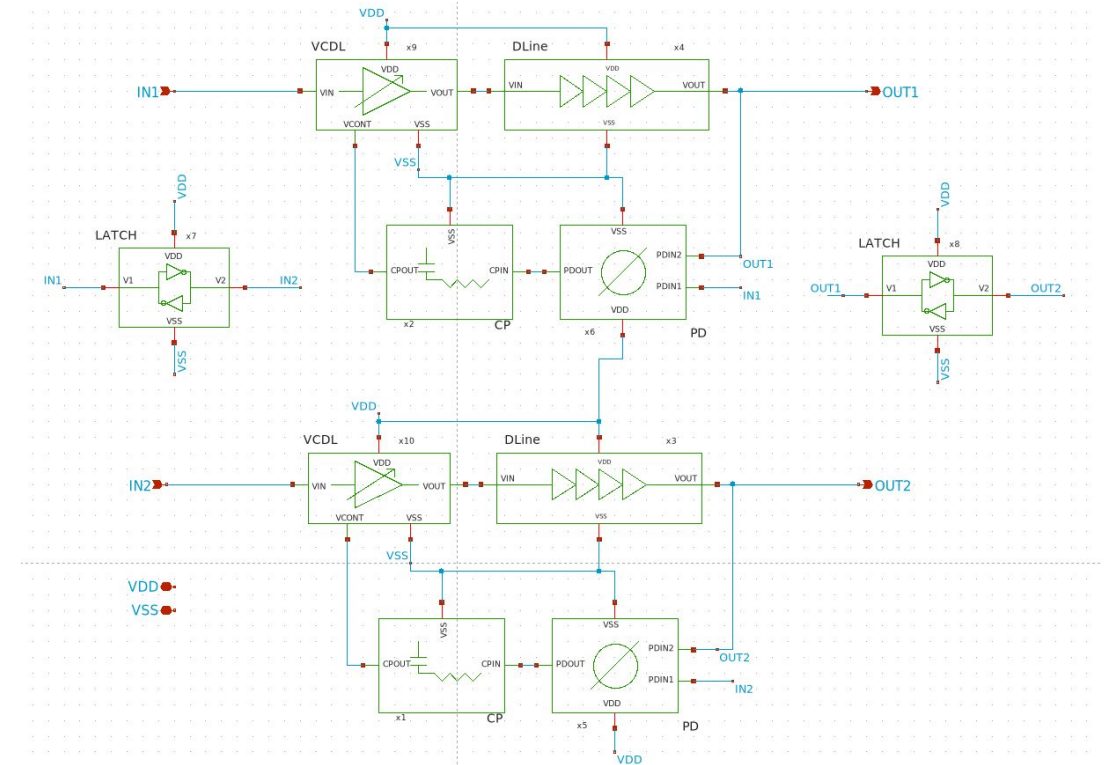
2

Output: *

2

Bidirectional: *

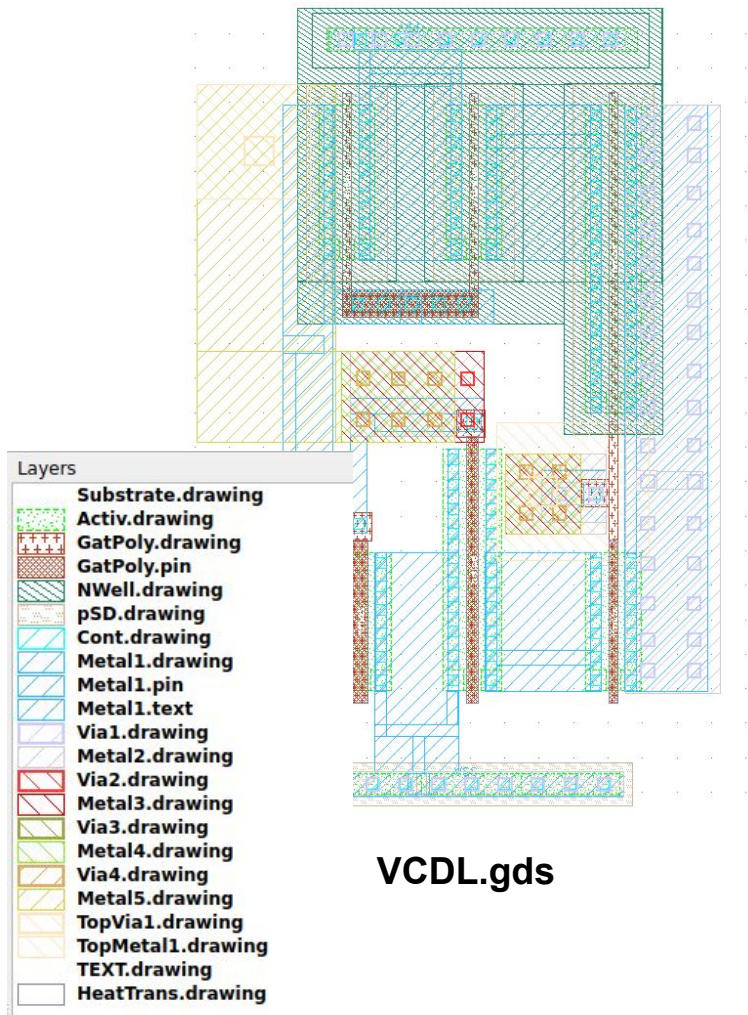
2



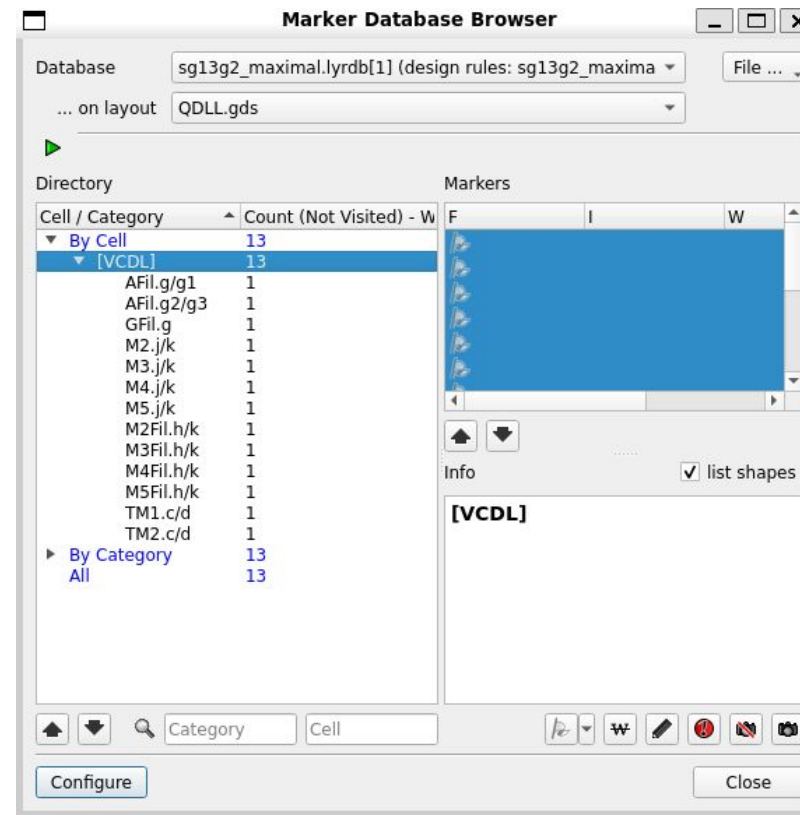
QDLL.sch

3 - QDLL Schematics & Layouts

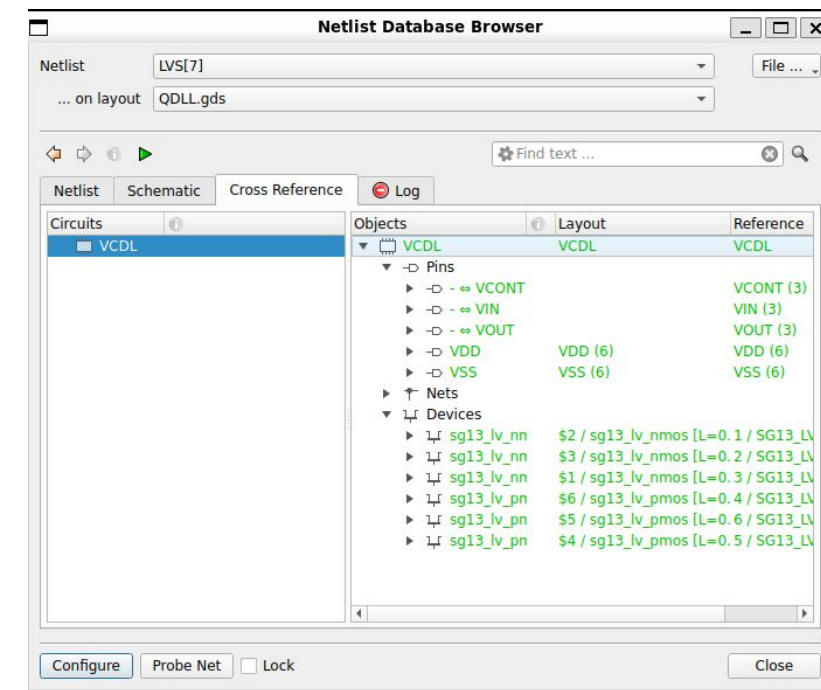
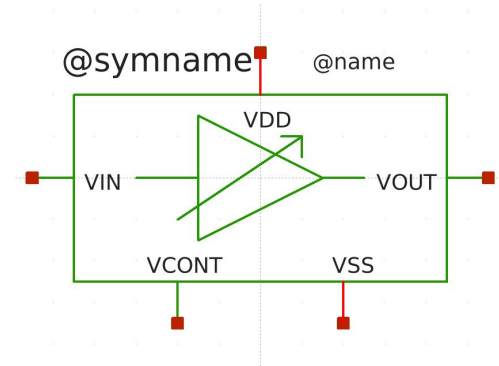
- Voltage-Controlled Delay Line (VCDL) :



VCDL.gds



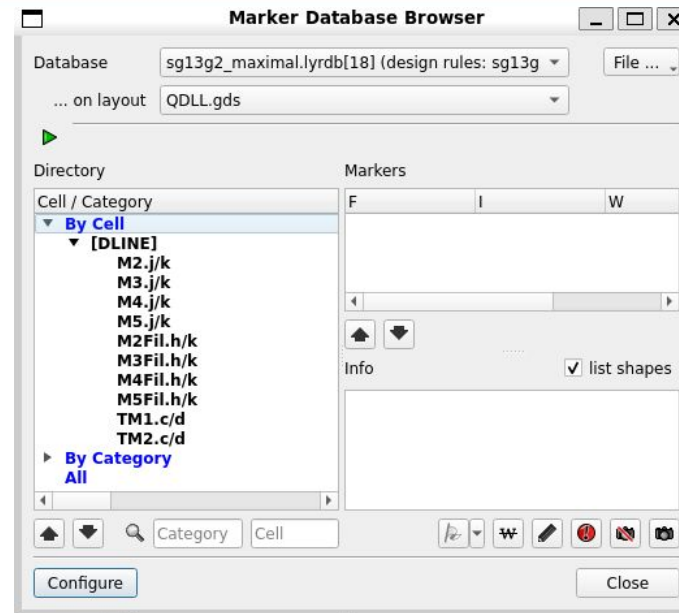
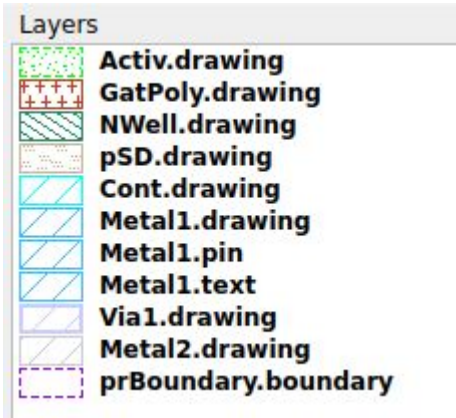
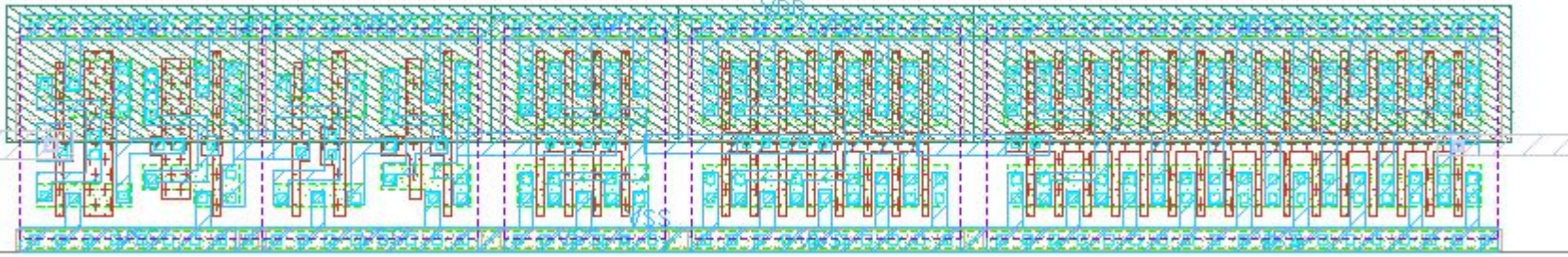
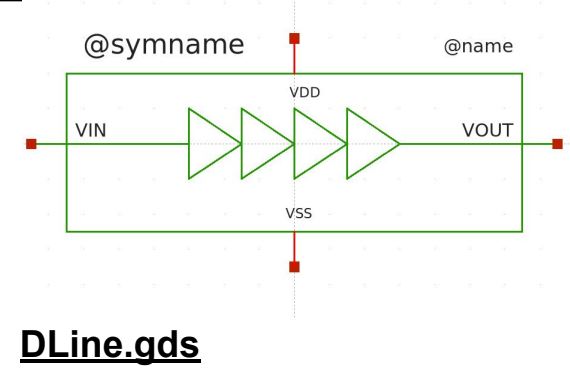
DRC clean (only density errors)



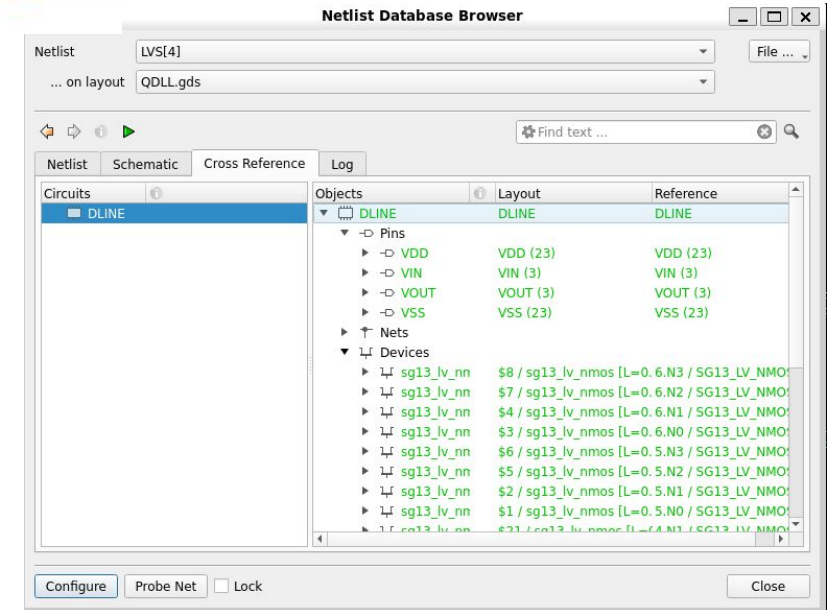
LVS Clean

3 - QDLL Schematics & Layouts

- Large delay Line + Buffer Stage (DLine) :



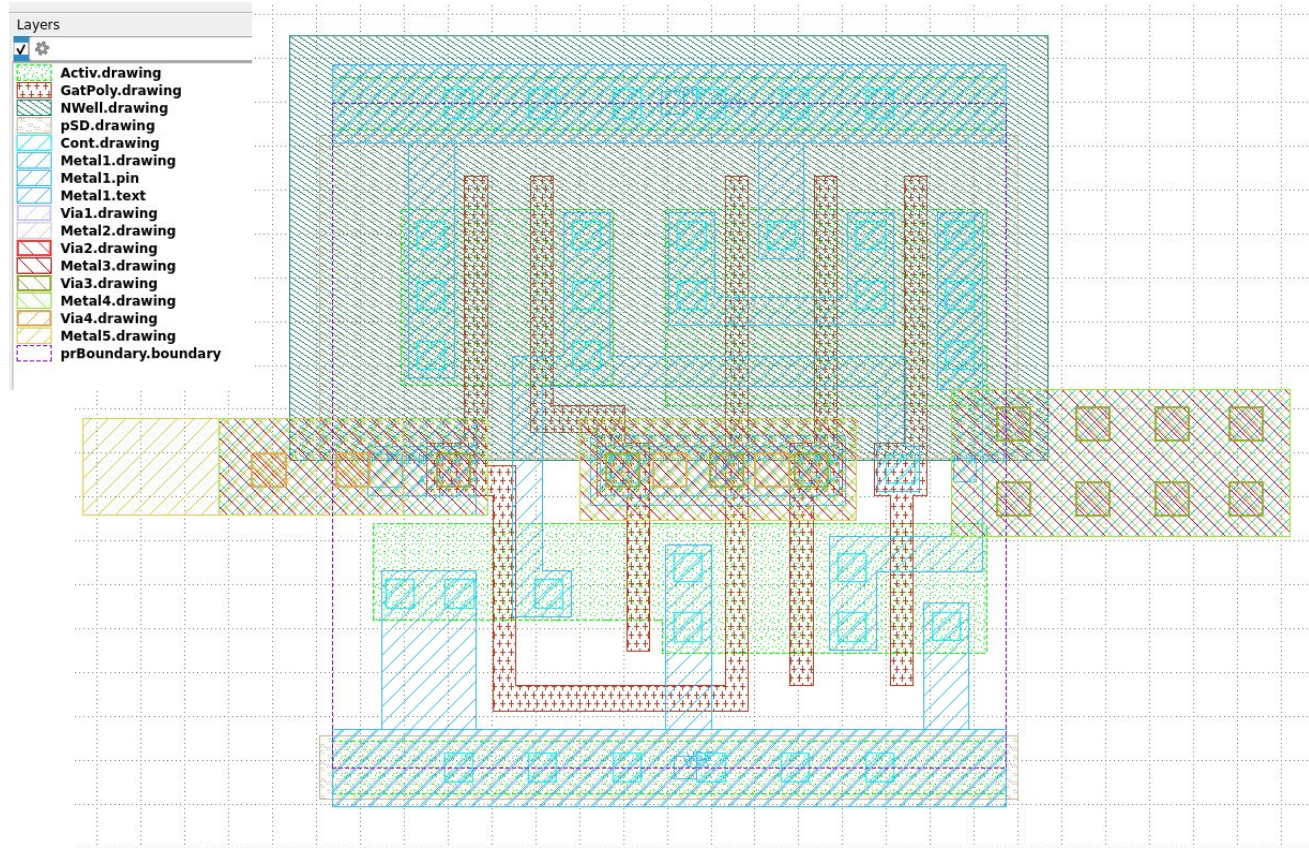
DRC clean (only density errors)



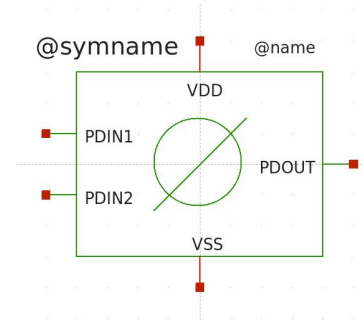
LVS Clean

3 - QDLL Schematics & Layouts

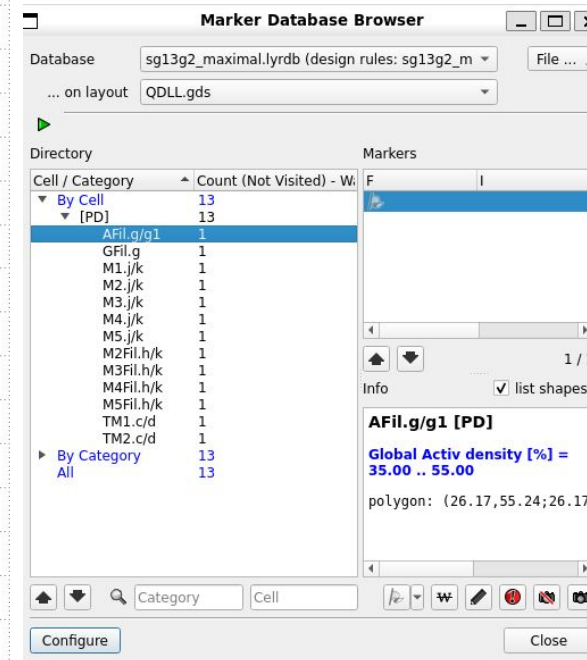
- Phase Detector (XOR cell) :



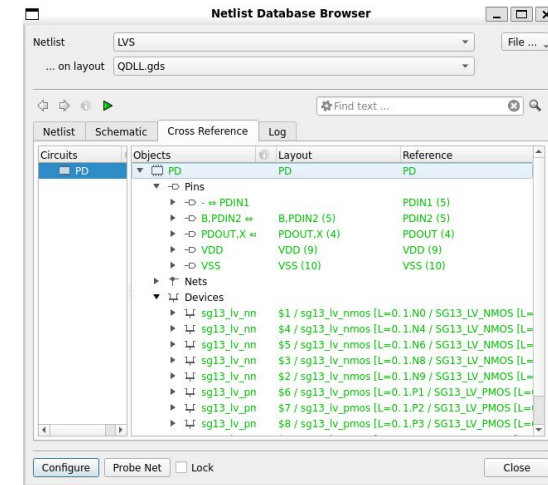
PD.gds



PD.sym



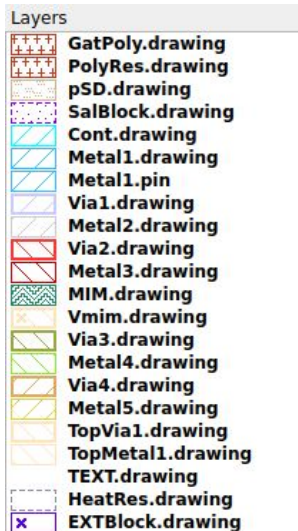
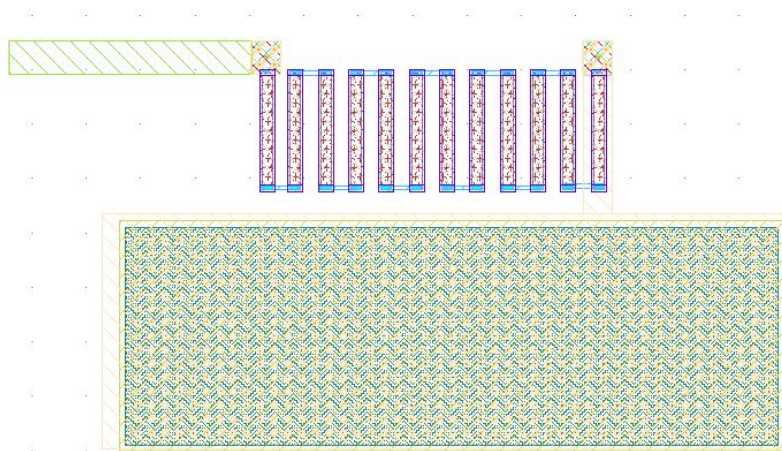
DRC clean (only density errors)



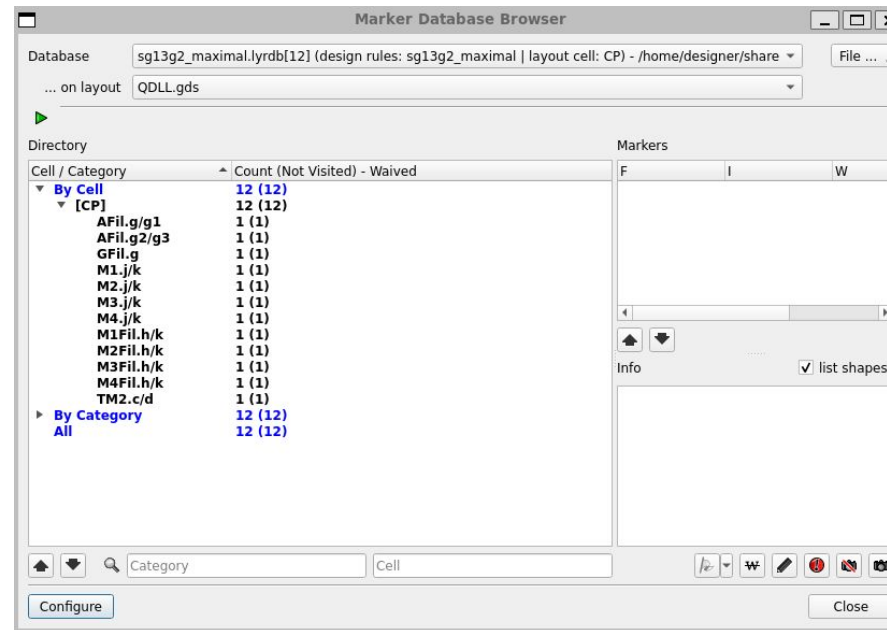
LVS Clean

3 - QDLL Schematics & Layouts

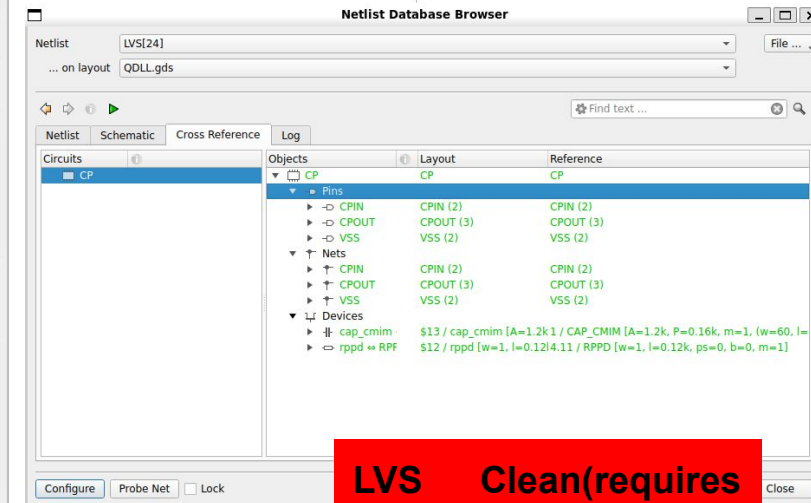
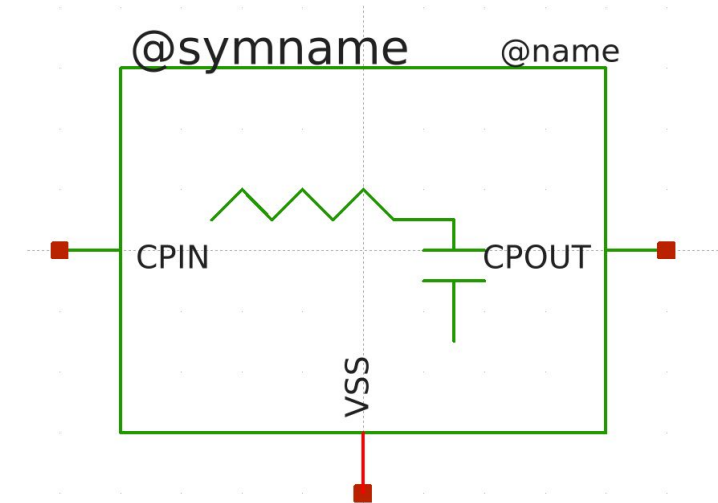
- Charge-Pump/ RC Loop Filter:



CP.gds



DRC clean (only density errors)

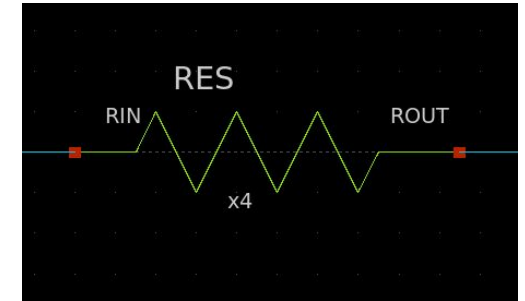


LVS Clean(requires an !sub connection with ptab cell to VSS on RES cell)

3 - QDLL Schematics & Layouts

- RESISTOR RES** important declaration:

To correctly pass the LVS, on the **RES** netlist, we neglected the indicated sub!'s sentences because the Klayout LVS Software identifies **sub!** as a Device, neglecting the **rppd** sentence.



Netlist	Schematic	Cross Reference	Log
Circuits	Objects	Layout	Reference
RES	RES	RES	RES
	Pins		
	RIN		RIN (2)
	ROUT		ROUT (2)
	SUB!		SUB! (2)
	VSS	VSS (2)	VSS (2)
	Nets		
	RIN		RIN (2)
	ROUT		ROUT (2)
	\$2	\$2 (1)	
	\$32	\$32 (1)	
	\$1 SUB!	\$1 (1)	SUB! (2)
	VSS	VSS (2)	VSS (2)
	Devices		
	SUB!	/	11 / SUB! [R=0, (L=0.12k, W=1)]
	ptap1 PTAP1	\$13 / ptap1 [A=3.6504, P=18.72]	14 / PTAP1 [(A=3.6504, P=18.72)]
	A	\$1 (1)	SUB! (2)
	C	VSS (2)	VSS (2)
	rppd	\$11 / rppd [w=1, l=0.12k, ps=0, b=0, m=1]	/
	rppd_1	\$32 (1)	
	rppd_2	\$2 (1)	

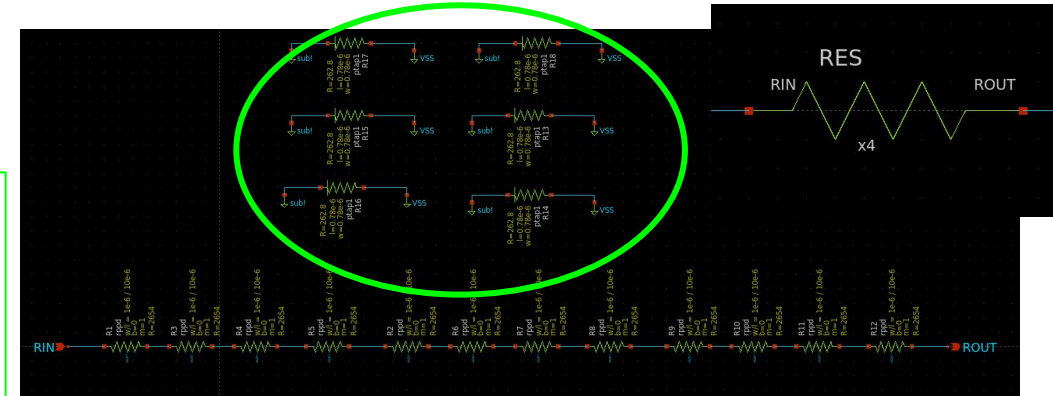


3 - QDLL Schematics & Layouts

- RESISTOR RES** important declaration:

To correctly pass the LVS, on the **RES** netlist, we neglected the indicated VSS's (body connections) sentences because the Klayout LVS Software identifies **VSS** as a Device, neglecting the **rppd** sentence.

To fix this we add custom **ptaps** to ensure the correctly body connection



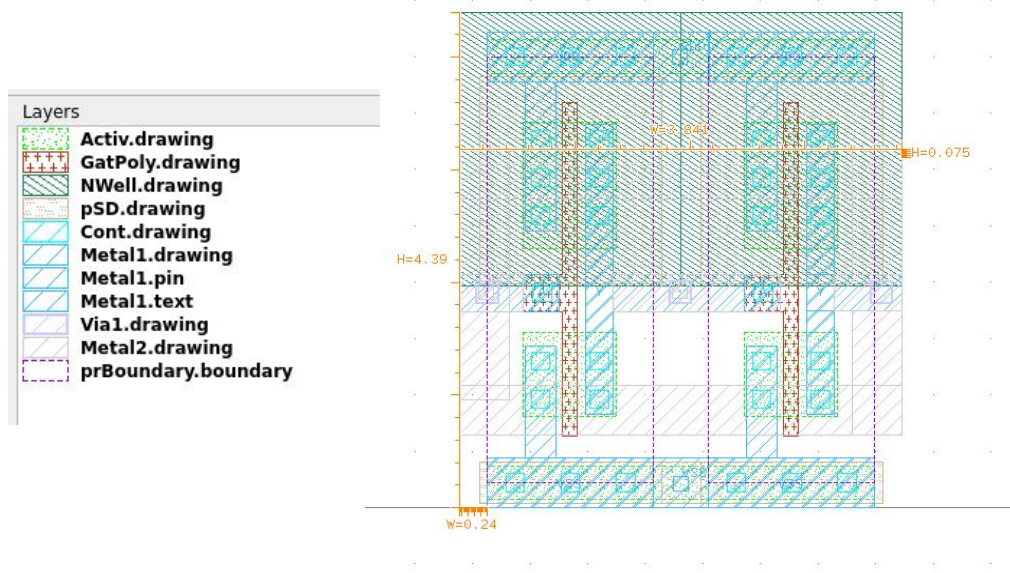
Netlist	Schematic	Cross Reference	Log
Circuits	Objects	Layout	Reference
RES	RES	RES	RES
Pins	Pins	Pins	Pins
RIN	RIN	RIN	RIN
ROUT	ROUT	ROUT	ROUT
SUB!	SUB!	SUB!	SUB!
VSS	VSS	VSS	VSS
Nets	Nets	Nets	Nets
\$2 ↔ RIN	\$2 ↔ RIN	\$2 (1)	RIN (2)
\$32 ↔ ROUT	\$32 ↔ ROUT	\$32 (1)	ROUT (2)
\$1 ↔ SUB!	\$1 ↔ SUB!	\$1 (1)	SUB! (2)
VSS	VSS	VSS (2)	VSS (2)
Devices	Devices	Devices	Devices
ptap1 ↔ PTAP1	ptap1 ↔ PTAP1	\$13 / ptap1 [A=3.6504, P=18.72]	14 / PTAP1 [(A=3.6504, P=18.72)]
A	A	\$1 (1)	SUB! (2)
C	C	VSS (2)	VSS (2)
rppd ↔ RPPD	rppd ↔ RPPD	\$11 / rppd [w=1, l=0.12k, ps=0, b=0, m=1]	11 / RPPD [w=1, l=0.12k, ps=0, b=0, m=1]
rppd_1 ↔ RPPD_1	rppd_1 ↔ RPPD_1	\$32 (1)	ROUT (2)
rppd_2 ↔ RPPD_2	rppd_2 ↔ RPPD_2	\$2 (1)	RIN (2)

Nets match, but the choice was ambiguous. This may lead to mismatching nets in other places.

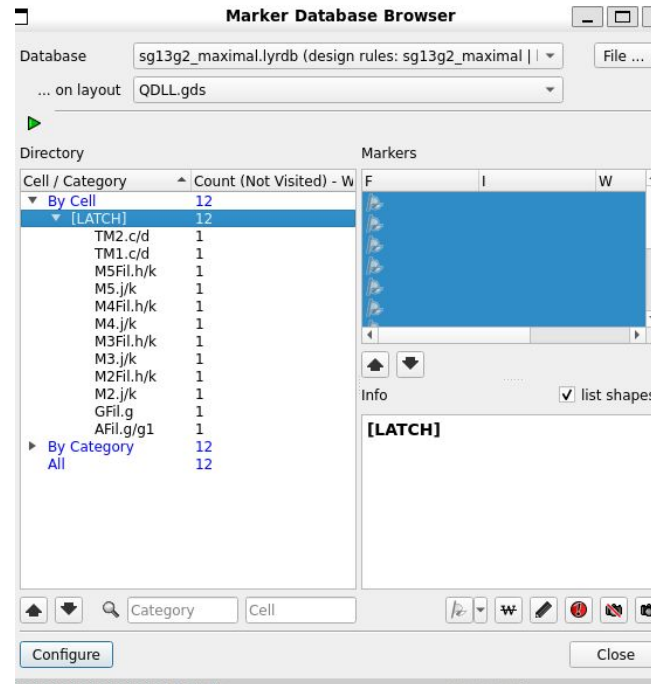
```
** sch_path: /home/designer/shared/2026/Cells/RES.s
.subckt RES ROUT RIN
*.PININFO RIN:I ROUT:O
R1 net1 RIN rppd w=1e-6 l=10e-6 m=1 b=0
R3 net2 net1 rppd w=1e-6 l=10e-6 m=1 b=0
R4 net3 net2 rppd w=1e-6 l=10e-6 m=1 b=0
R5 net4 net3 rppd w=1e-6 l=10e-6 m=1 b=0
R2 net5 net4 rppd w=1e-6 l=10e-6 m=1 b=0
R6 net6 net5 rppd w=1e-6 l=10e-6 m=1 b=0
R7 net7 net6 rppd w=1e-6 l=10e-6 m=1 b=0
R8 net8 net7 rppd w=1e-6 l=10e-6 m=1 b=0
R9 net9 net8 rppd w=1e-6 l=10e-6 m=1 b=0
R10 net10 net9 rppd w=1e-6 l=10e-6 m=1 b=0
R11 net11 net10 rppd w=1e-6 l=10e-6 m=1 b=0
R12 ROUT net11 rppd w=1e-6 l=10e-6 m=1 b=0
.ends
```


3 - QDLL Schematics

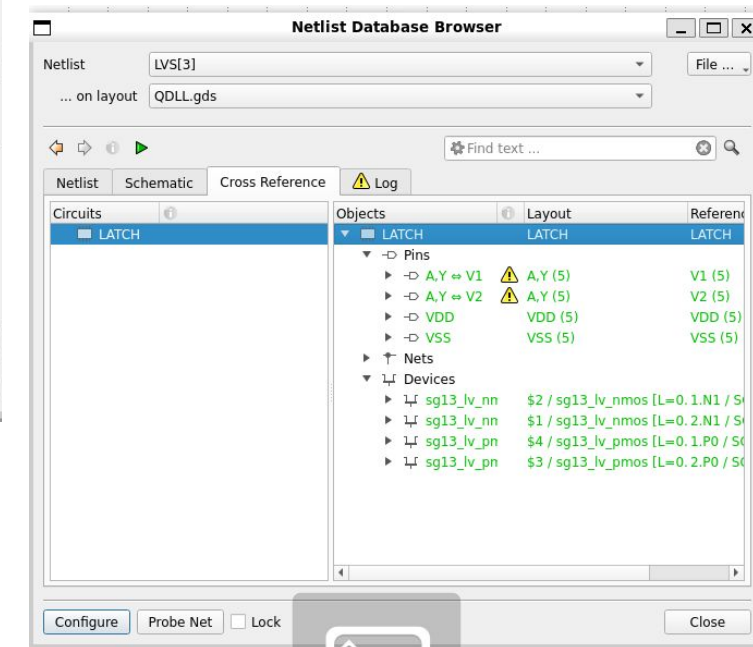
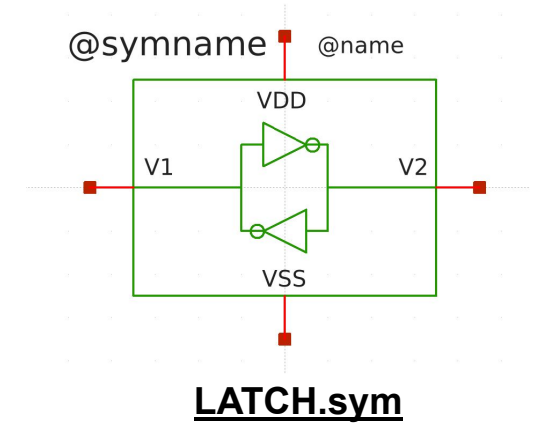
- Flywheel Latches:



LATCH.gds



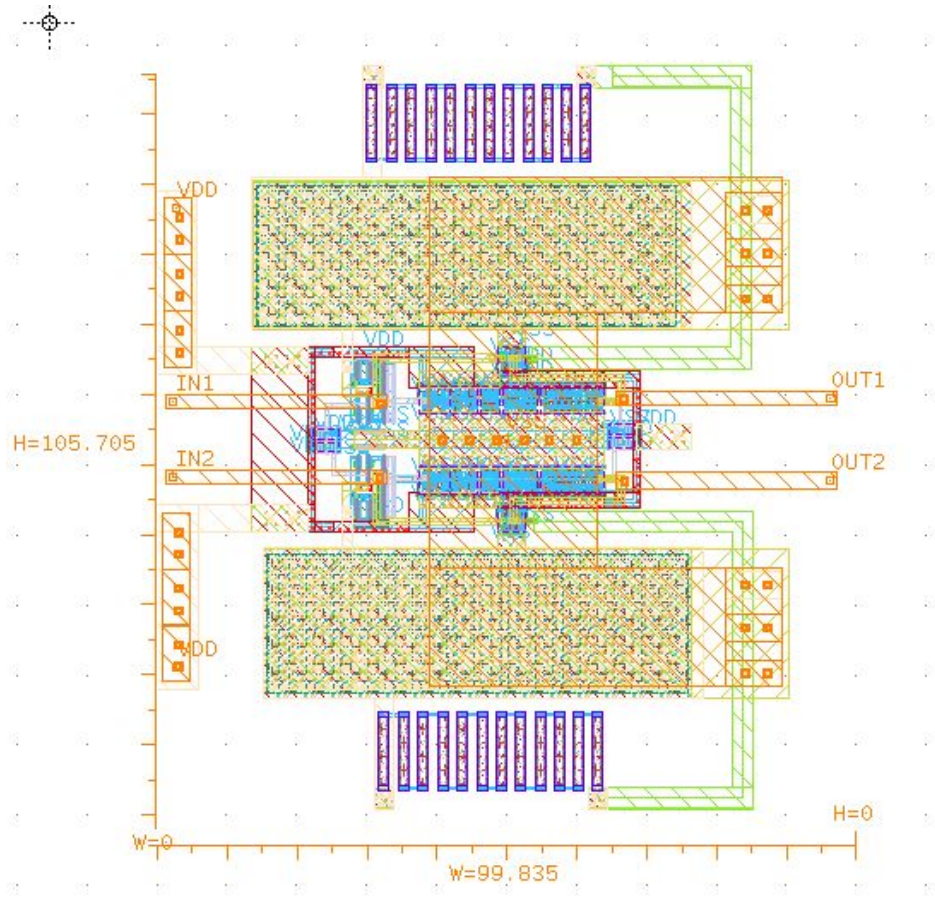
DRC clean (only density errors)



LVS Clean

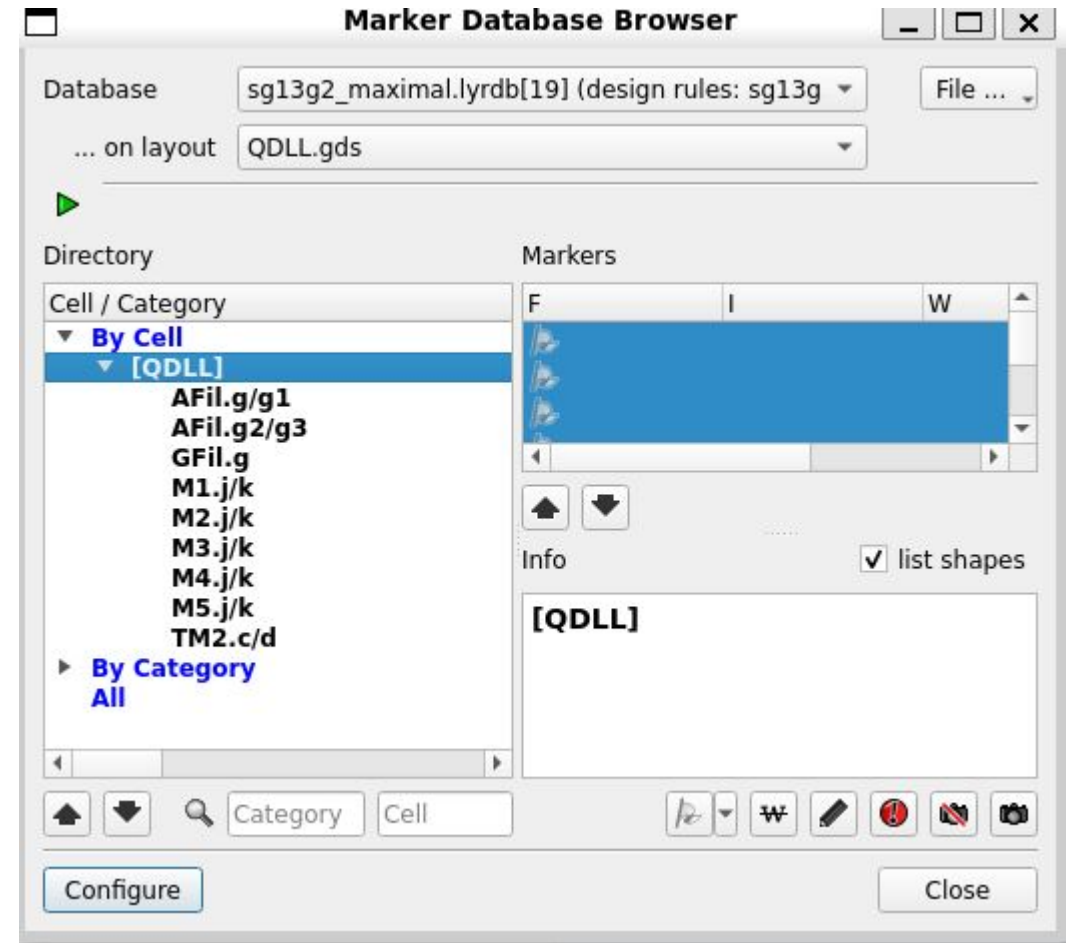
4- QDLL TOP Layout

QDLL.gds Top Layout Verification



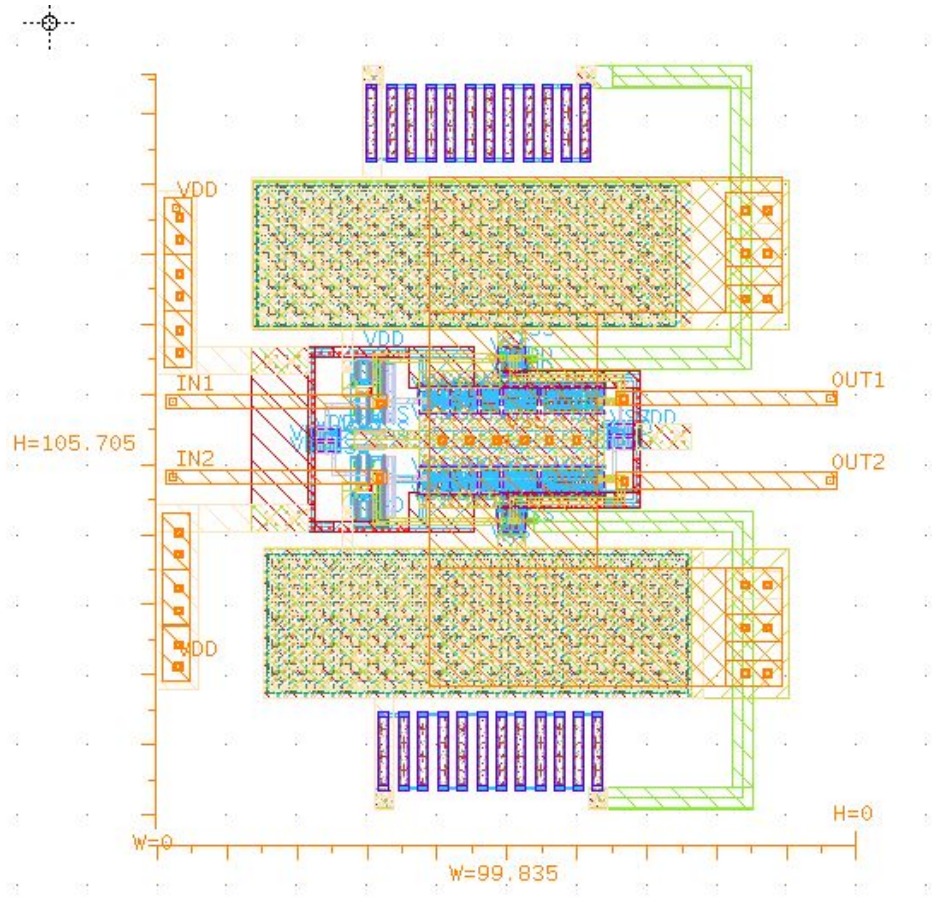
Vertical View

Area aprox: $100\mu\text{m} \times 100\mu\text{m} = 0.1\mu\text{m}^2 =$



**DRC Clean (ONLY DENSITY
ERRORS, SOLVING ON
INTEGRATION STAGE)**

QDLL.gds Top Layout Verification



Vertical View

Area aprox: $100\mu\text{m} \times 100\mu\text{m} = 0.1\mu\text{m}^2 =$

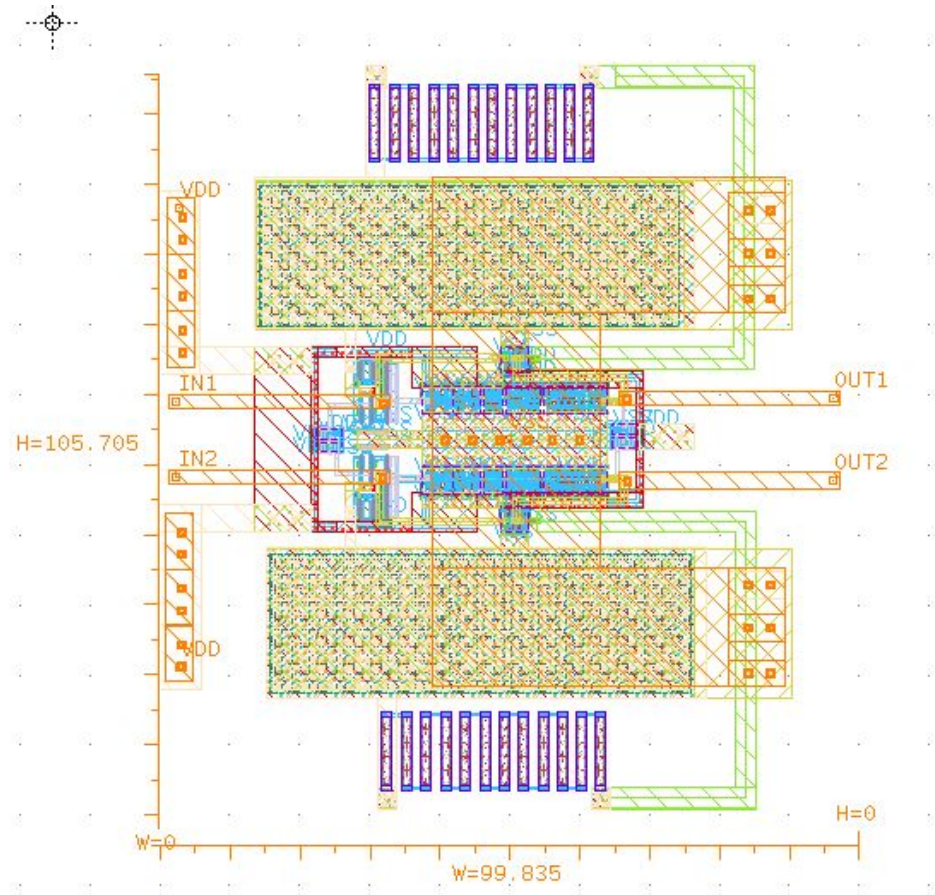
Netlist: LVS[22]
... on layout: QDLL.gds

Netlist Schematic Cross Reference Log

Circuits	Objects	Layout	Reference
QDLL	QDLL	QDLL	QDLL
	Pins		
	IN1	IN1 (11)	IN1 (11)
	IN2	IN2 (11)	IN2 (11)
	OUT1	OUT1 (11)	OUT1 (11)
	OUT2	OUT2 (11)	OUT2 (11)
	VDD	VDD (79)	VDD (79)
	VSS	VSS (83)	VSS (83)
	Nets		
	\$55 ↔ 1.NE	\$55 (4)	1.NET1 (4)
	\$73 ↔ 1.NE	\$73 (2)	1.NET2 (2)
	\$56 ↔ 1.NE	\$56 (2)	1.NET3 (2)
	\$61 ↔ 2.5.I	\$61 (4)	2.5.NET1 (4)
	\$62 ↔ 2.5.I	\$62 (4)	2.5.NET2 (4)
	\$63 ↔ 2.5.I	\$63 (4)	2.5.NET3 (4)
	\$65 ↔ 2.6.I	\$65 (4)	2.6.NET1 (4)
	\$66 ↔ 2.6.I	\$66 (4)	2.6.NET2 (4)
	\$67 ↔ 2.6.I	\$67 (4)	2.6.NET3 (4)
	\$69 ↔ 2.NE	\$69 (4)	2.NET1 (4)
	\$70 ↔ 2.NE	\$70 (4)	2.NET2 (4)
	\$64 ↔ 2.NE	\$64 (4)	2.NET3 (4)
	\$68 ↔ 2.NE	\$68 (4)	2.NET4 (4)
	\$77 ↔ 4.1.I	\$77 (5)	4.1.NET1 (5)
	\$81 ↔ 4.1.I	\$81 (2)	4.1.NET3 (2)
	\$78 ↔ 4.1.I	\$78 (3)	4.1.NET5 (3)
	\$80 ↔ 4.1.I	\$80 (2)	4.1.NET6 (2)
	\$31 ↔ 7.NE	\$31 (4)	7.NET1 (4)
	\$32 ↔ 7.NE	\$32 (2)	7.NET2 (2)
	\$50 ↔ 7.NE	\$50 (2)	7.NET3 (2)
	\$39 ↔ 8.5.I	\$39 (4)	8.5.NET1 (4)
	\$40 ↔ 8.5.I	\$40 (4)	8.5.NET2 (4)
	\$41 ↔ 8.5.I	\$41 (4)	8.5.NET3 (4)
	\$43 ↔ 8.6.I	\$43 (4)	8.6.NET1 (4)
	\$44 ↔ 8.6.I	\$44 (4)	8.6.NET2 (4)
	\$45 ↔ 8.6.I	\$45 (4)	8.6.NET3 (4)
	\$47 ↔ 8.NE	\$47 (4)	8.NET1 (4)
	\$48 ↔ 8.NE	\$48 (4)	8.NET2 (4)
	\$42 ↔ 8.NE	\$42 (4)	8.NET3 (4)
	\$46 ↔ 8.NE	\$46 (4)	8.NET4 (4)
	\$28 ↔ 9.1.I	\$28 (5)	9.1.NET1 (5)
	\$29 ↔ 9.1.I	\$29 (2)	9.1.NET3 (2)
	\$34 ↔ 9.1.I	\$34 (3)	9.1.NET5 (3)

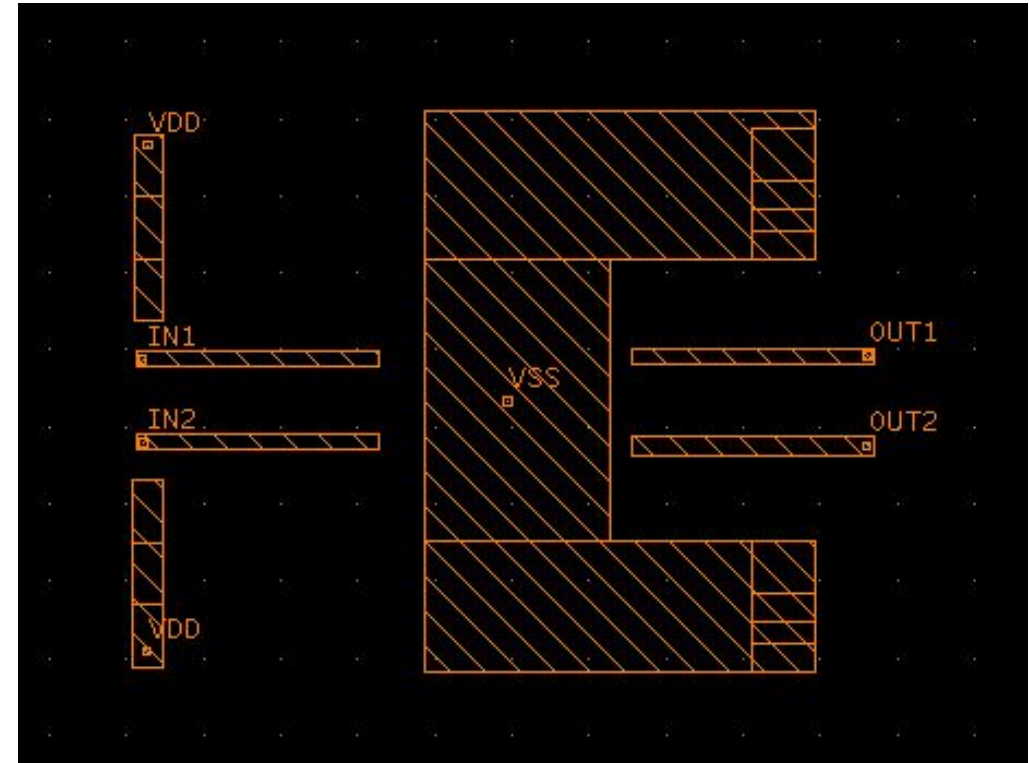
LVS Clean

QDLL.gds Top Layout CONNECTION



Vertical View

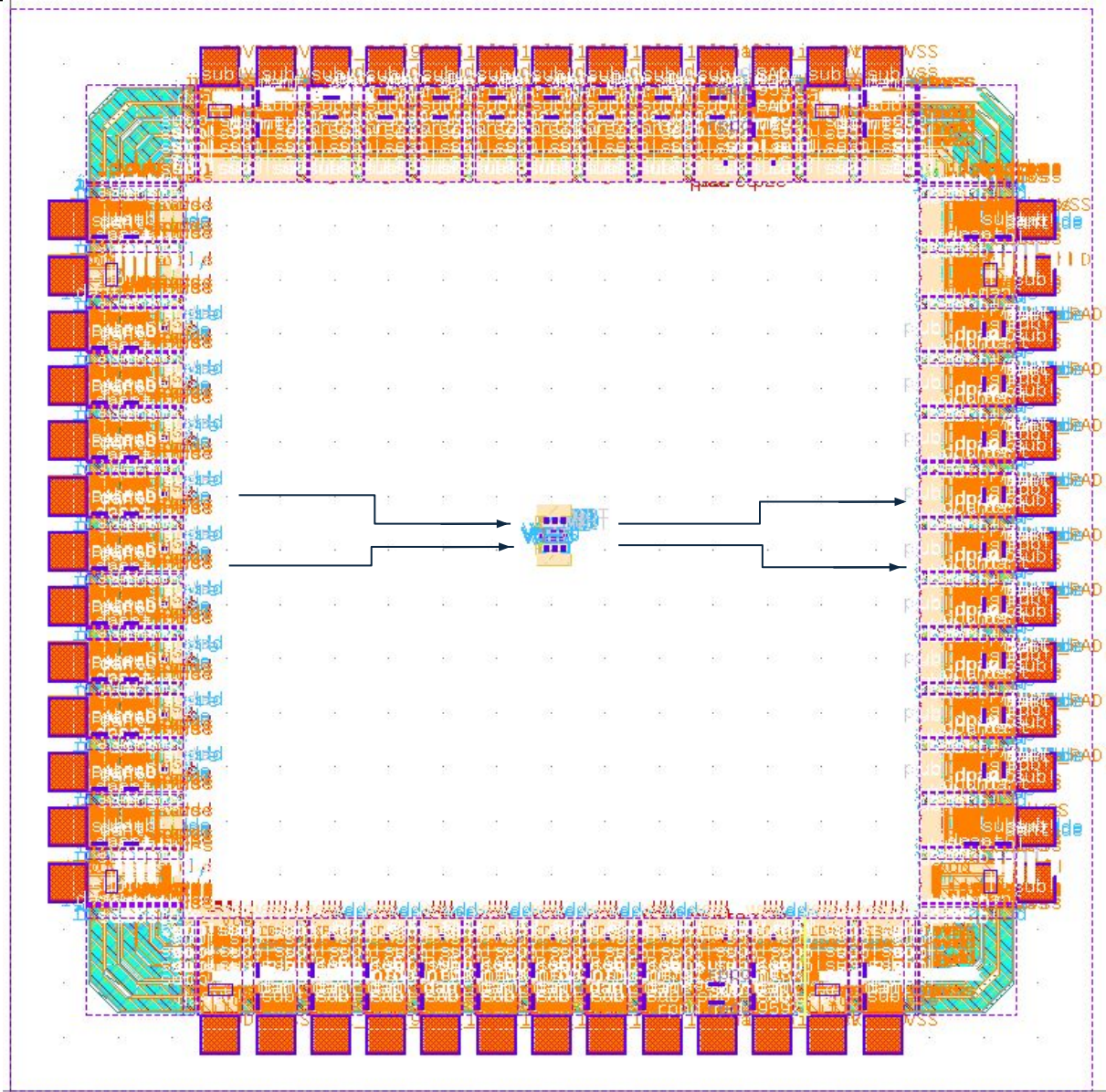
Area aprox: $100\mu\text{m} \times 100\mu\text{m} = 0.01\text{mm}^2$



TOP METAL 2 CONNECTIONS:

- **POWER:** VDD, VSS
- **INPUTS:** IN1, IN2
- **OUTPUT:** OUT1, OUT2

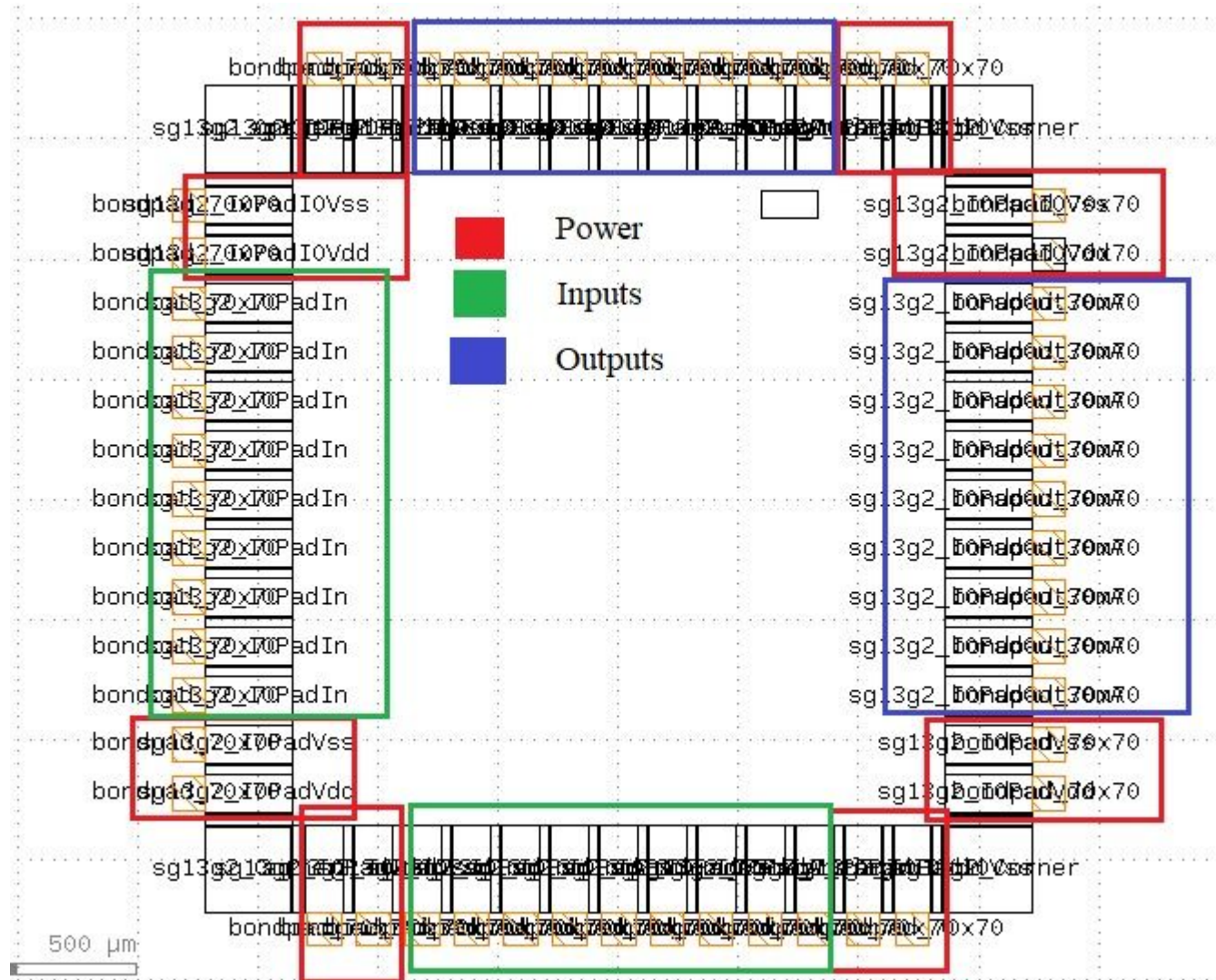
QDLL Top Layout Placement v1



One possible goal would be to place the QDLL vertically and connect the inputs on the left and the outputs on the right with high metal top routing.

For a 250 MHz signal, standard IO cells have been tested and found to perform well. However, if available, we prefer to interface with 50 Ω systems using IO cells with matched impedance.

QDLL layout within the pad ring



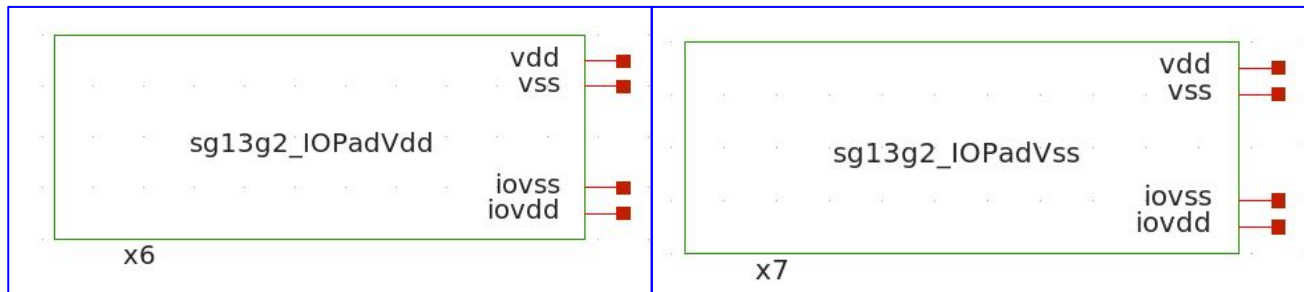
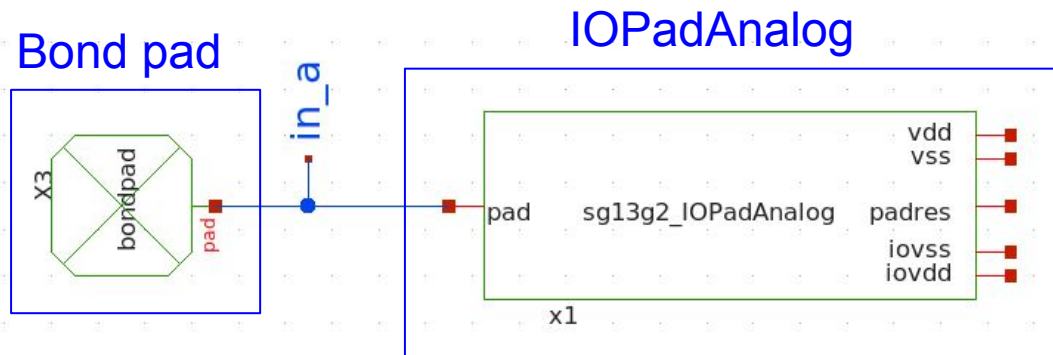
5 - QDLL Pins and IO CELLS

Pins and IO Cells

QDLL Pins

N° Pins: 4

- 2 Input Pins
- 2 Output Pin
- 2 Power Supply Pins (VDD , VSS)



Pin Type	Name	IOCELL
Input	inpad1	sg13g2_IOPADAnalog
Input	inpad2	sg13g2_IOPADAnalog
Output	outpad1	sg13g2_IOPADAnalog
Output	outpad1	sg13g2_IOPADAnalog
Power Supply	VDD	sg13g2_IOPadVdd
Power Supply	VSS	sg13g2_IOPadVss

Power Supply IOPads

6- QDLL Project Status

Project Status

Project Status by Block:

TOP	Sub-top	Sub-Block	Symbols & Schematics	Testbench	Prelayout results	IOCELLS results (TT)	IOCELLS results (SS-FF)	Layout (DRC + LVS)	Postlayout results
QDLL TOP	-	-	Done	Done	Done	Done	W/P	Done	W/P
-	SE_QDLL	-	Done	Done	Done	Done	WIP	Done	WIP
-	-	VCDL	Done	Done	Done	Done	W/P	Done	W/P
-	-	DLine	Done	Done	Done	Done	WIP	Done	WIP
-	-	CP	Done	Done	Done	Done	WIP	Done	WIP
-	-	PD	Done	Done	Done	Done	WIP	Done	WIP
-	LATCH	-	Done	Done	Done	Done	WIP	Done	WIP

For this delivering, we were working on the DRC (minimum and maximum) verification of all cells to achieve the desired LVS verification. Next, we plan to compare the extracted views using Magic in order to obtain more realistic results.

On the other hand, multi-corner simulation results are desirable for our project, and we are actively working toward that goal.